Week #9: [Knee]

[Knee Joint: Osteology and Ligamentous Anatomy]
- Knee osteology
- Knee ligaments and stability
- Function of ACL/PCL

[Ligamentous Injuries of the Knee] (ACL and PCL injuries, MCL and LCL injuries, Evaluation, treatment, and return to play decisions) ***No text notes from Dr. Briscoe**

[Meniscus Anatomy and Function]
- Anatomy and function of meniscus
- Bucket-handle tears
- Locking mechanism of the meniscus

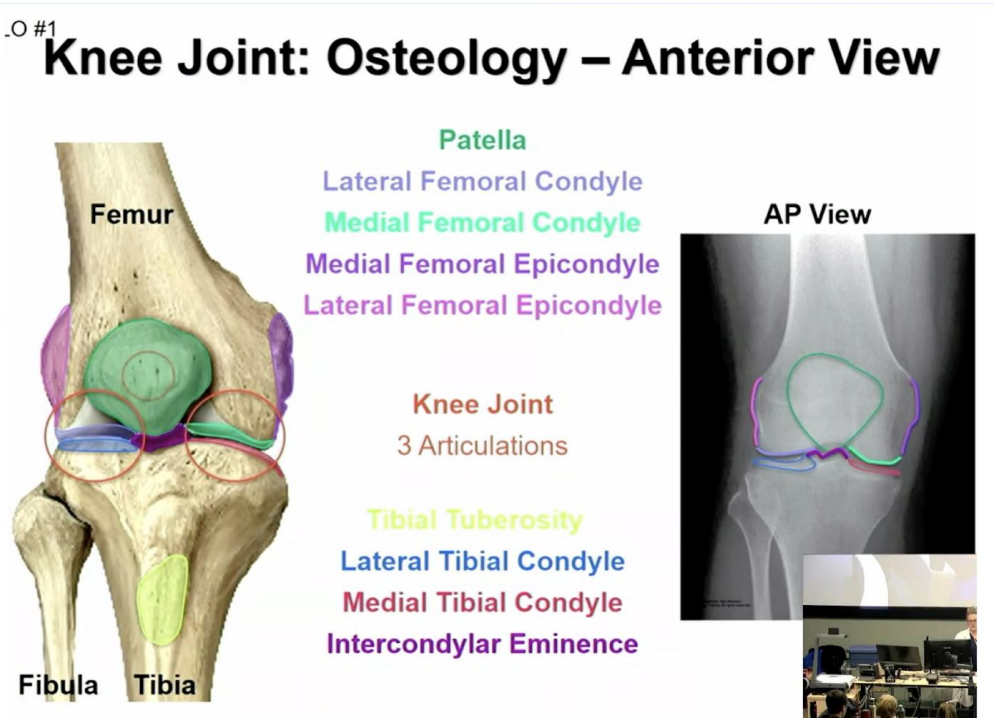
[Meniscus Injuries] (Pathology, mechanism of injury, symptoms, Treatment of meniscal injuries, Unhappy triad) ***No text notes from Dr. Briscoe**

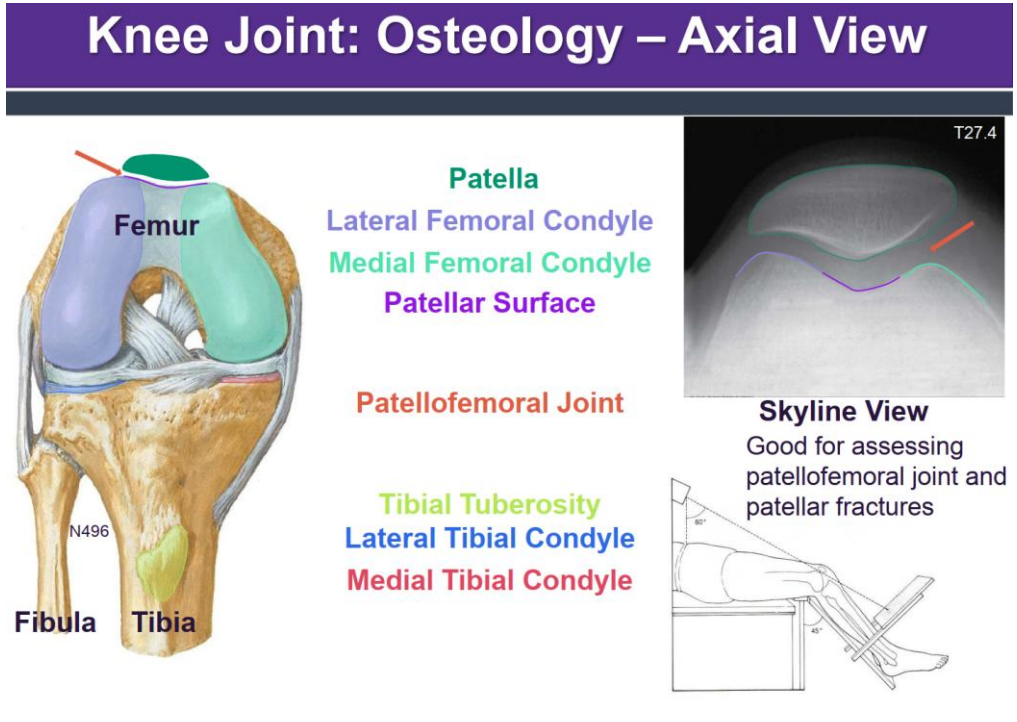
[Anterior Knee Pain Anatomy]
- Anatomy of patellofemoral joint
- Stability forces of patellofemoral joint

[Anterior Knee Pain Pathology] (Patellofemoral Pain Syndrome (PFPS) - Symptoms, Presentation, Treatment) ***No text notes from Dr. Briscoe**

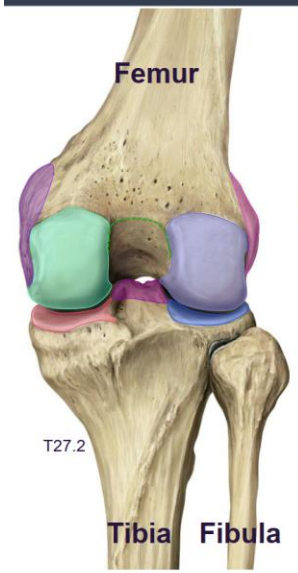
[MISC Knee Structures]
- Anatomy of Knee Bursae

Knee Joint: Osteology and Ligamentous Anatomy
- Knee osteology
- Knee ligaments and stability
- Function of ACL/PCL



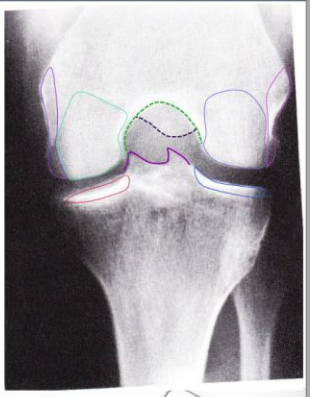


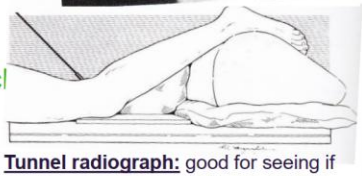
Knee Joint: Osteology – Posterior View



Patella
Lateral Femoral Condyle
Medial Femoral Condyle
Medial Femoral Epicondyle
Lateral Femoral Epicondyle

Intercondylar Eminence
Lateral Tibial Condyle
Medial Tibial Condyle
Intercondylar Fossa (Notch)





Tunnel radiograph: good for seeing if any loose body or bone tissue in joint cavity

Knee Joint: Osteology – Lateral/Sagittal View



Patella
Lateral Femoral Condyle
Lateral Tibial Condyle
Tibial Tuberosity

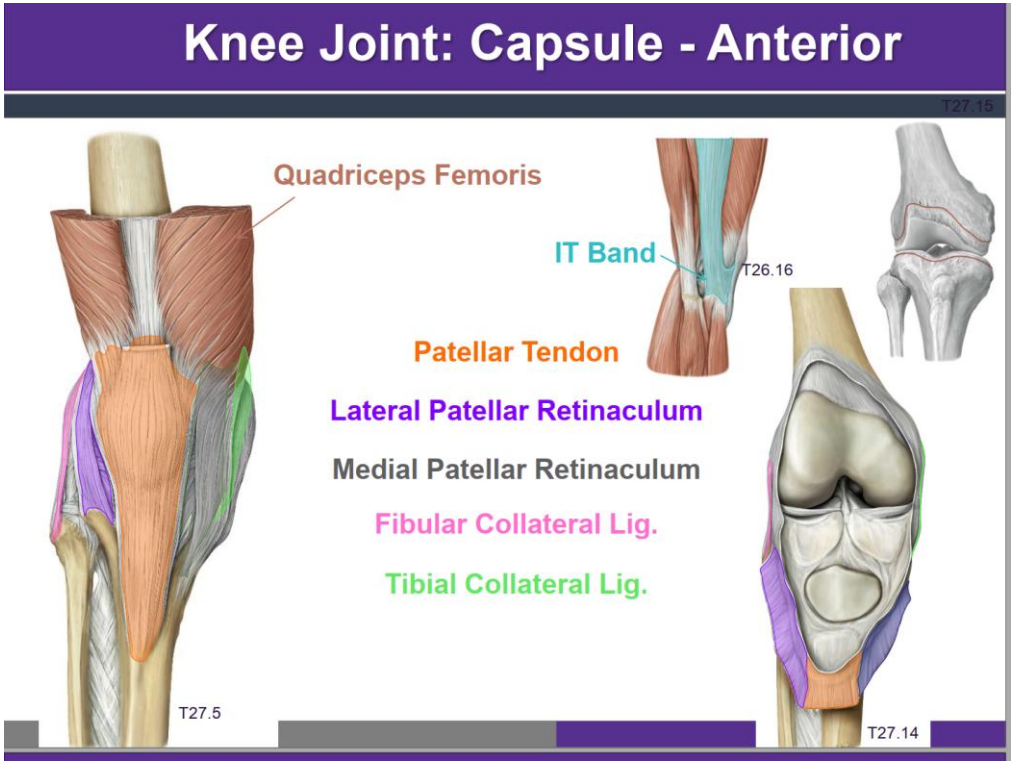


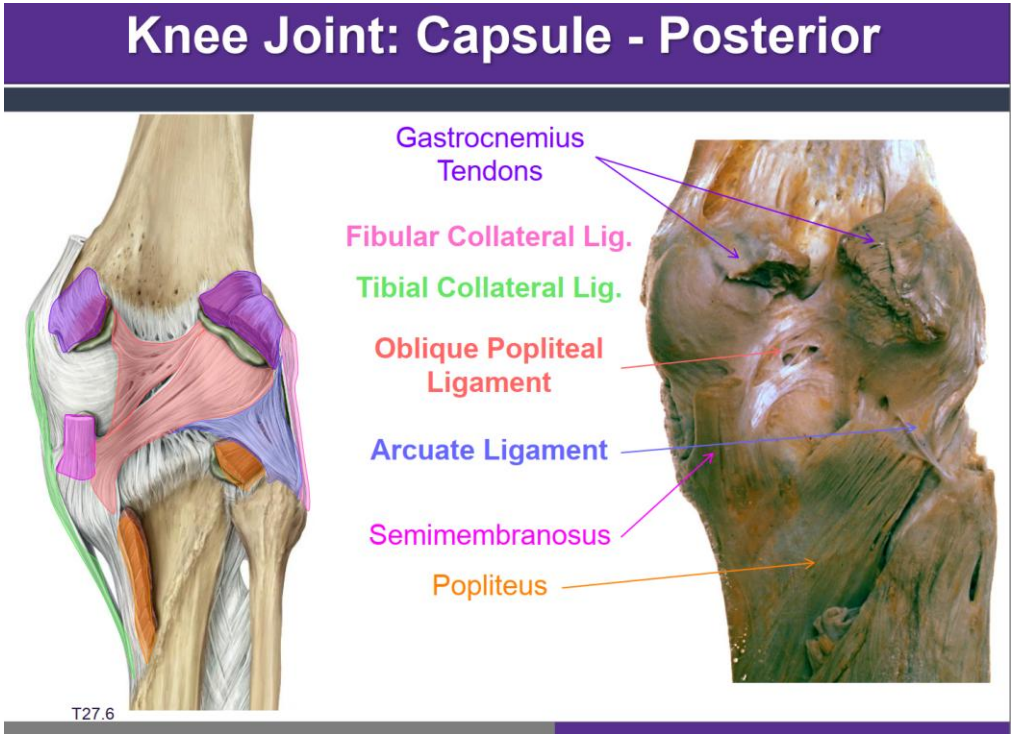
I. Knee Joint

a. **Review:** osteology of the distal femur and proximal tibia
   i. ***Type:*** **Hinge** synovial joint; also allows for some **gliding**/**rotation** about the long axis
   ii. ***Articulation***: large **lateral and medial femoral condyles** articulate with **lateral and medial tibial condyles; femur** articulates with **patella**

Supporting Structures:

a. Stability is primarily facilitated by the ***surrounding musculature*** (especially the **quadriceps femoris**), with added stability conveyed by **ligamentous support**





Largest joint in the human body with the **largest joint capsule** and **extensive synovial membrane**

a. **Anteriorly:** the joint capsule is reinforced (and fibrous joint capsule is actually replaced) by the **tendon of the quadriceps femoris/patellar ligament** and by aponeurotic extensions of the vastus lateralis and the vastus medialis (medial and lateral patellar retinacula).
b. **Medially:** the joint capsule is reinforced by strong, long, and broad **tibial collateral ligament (TCL/MCL)** with muscular support from the **pes anserinus** (SGT), the **semimembranosus,** and the medial head of the **gastrocnemius**. Deep fibers of the TCL are firmly attached to the medial meniscus.

Clinical Note: TCL Injury: Frequency of TCL injuries are greater than fibular collateral ligament injuries. Common mechanisms include a lateral (valgus) force to an extended knee or excessive twisting/rotation of the flexed knee and because the medial meniscus is attached to the TCL, if the TCL is damaged the medial meniscus is also susceptible to damage.

- **TCL Assessment:** Testing of TCL integrity is performed by subjecting the flexed knee to a valgus stress (i.e. forced abduction of the leg at the femur), greater than normal abduction indicates TCL damage.

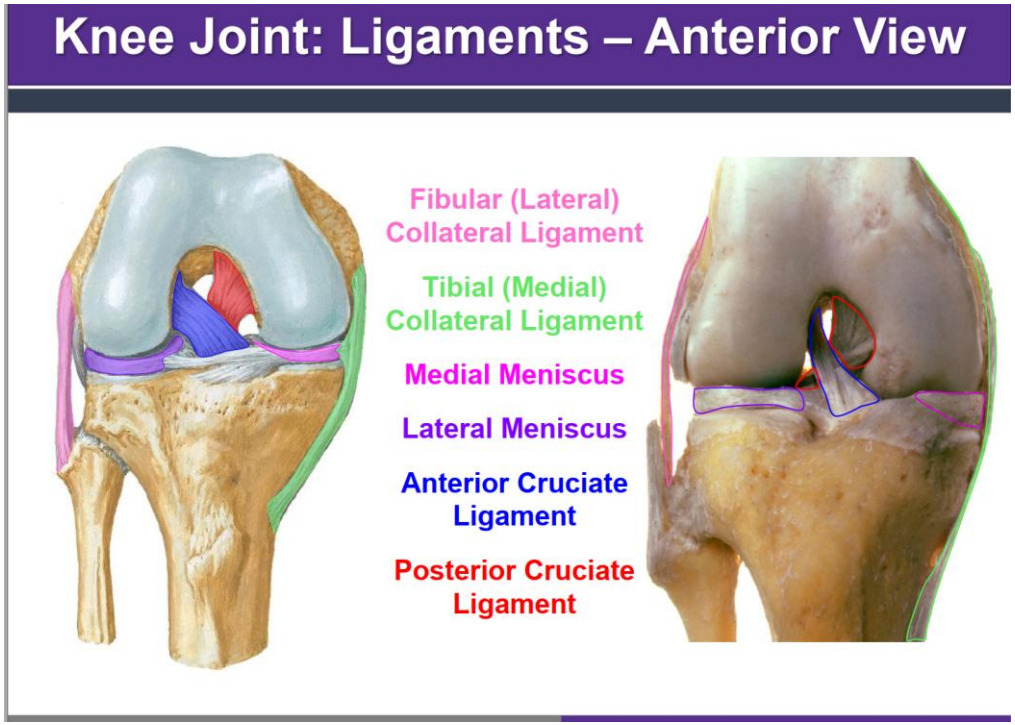
Laterally: joint capsule is reinforced by the strong, thin, and cord-like fibular collateral ligament (FCL/LCL) with musculotendinous support from the biceps femoris and the iliotibial band

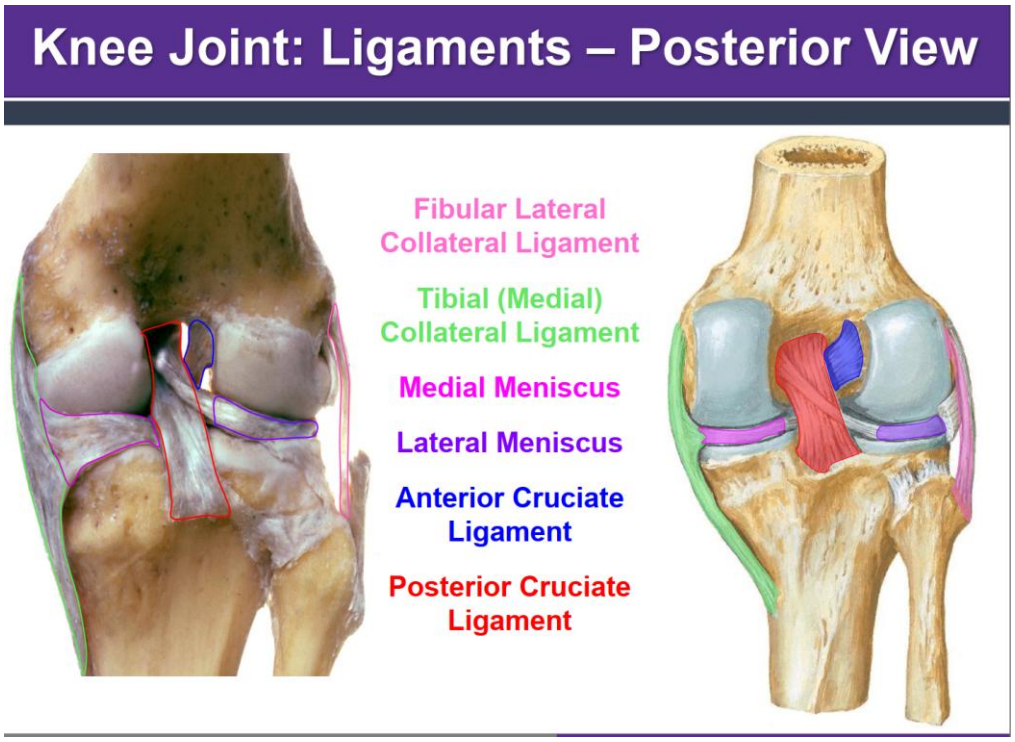
Clinical Note: FCL Injury: FCL can be damaged by a strong force applied on the medial aspect (varus) of the knee.

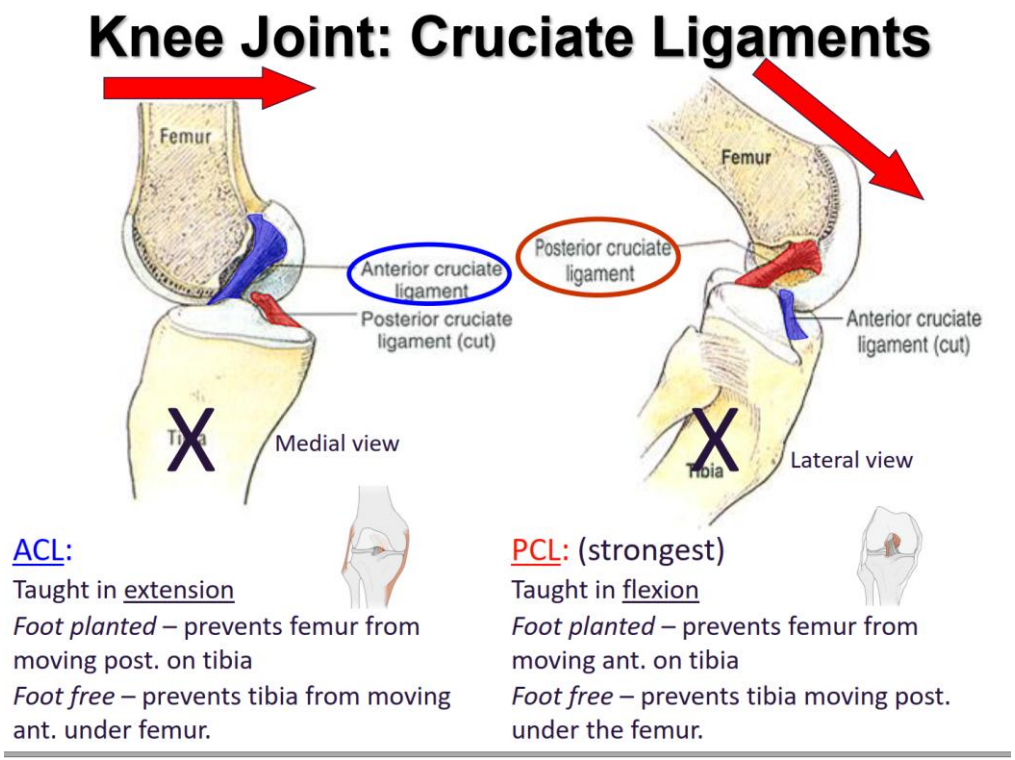
- **FCL Assessment:** Testing of the FCL is performed by subjecting the flexed knee to a varus stress (i.e. forced adduction of the leg at the femur), greater than normal adduction indicates FCL damage.

Posteriorly: the joint capsule covers the knee joint like a true fibrous capsule and is reinforce by the **oblique popliteal ligament**, which is a fibrous expansion from the **semimembranosus** tendon. The **arcuate popliteal ligament** arises from the posterior aspect of the fibular head, passes superomedially over the tendon of **popliteus,** and

spreads over the posterior surface of the knee.







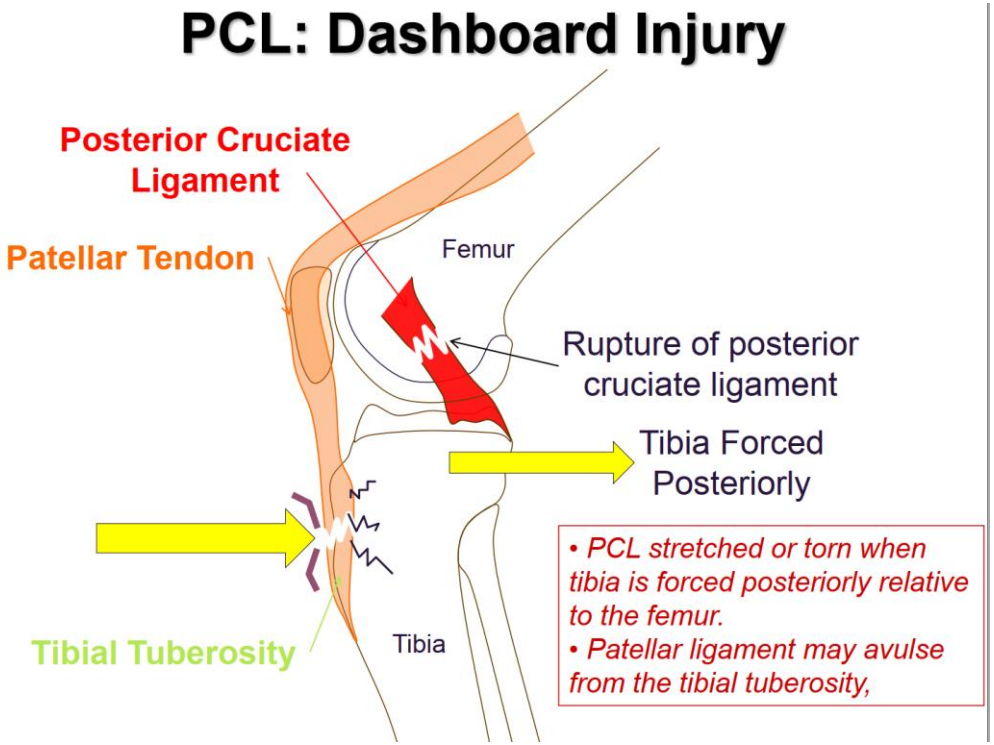
e. Internally: the cruciate ligaments and the menisci are located within the joint capsule.

- **Note:** The cruciate ligaments are named for where they are attached to the tibial plateau.

Anterior cruciate ligament (ACL): thinner/weaker of the two cruciate ligaments; courses inferomedially from the medial aspect of the lateral femoral condyle to the intercondylar fossa of the tibia (just anterior to the intercondylar eminence); prevents excessive anterior translation of the tibia under the femur (prevents hyperextension of knee joint) and resists excessive medial rotation of tibia relative to femur

Clinical Note: ACL injury: common mechanism of injury involves receiving **forceful impact to thigh**, when the foot is fixed, forcing the tibia to translate anteriorly relative to femur. Can also be ruptured when a force applied to a flexed knee causes excessive rotation and medial (valgus) stress (causing the "**Unhappy triad**" of injuries, in which the ACL, TCL, and medial meniscus are all damaged)

- **ACL assessment:** Testing of the ACL involves the examiner assessing the extent/degree of anterior translation of the tibia under the femur. Greater than normal anterior translation of the tibia is indicative of an ACL injury.
  - **Lachman test:** supine patient flexes knee ~300 and examiner draws tibia forward; considered gold standard for ACL testing (ACL function is better isolated with this test)
  - **Anterior drawer test:** supine patient flexes hip 450 and knee 90 0; examiner palpates the hamstring tendons to ensure they are relaxed, before drawing tibia forward



Posterior cruciate ligament (PCL): broader/stronger of the two cruciate ligaments; courses inferolaterally from the lateral side of the medial femoral condyle to the posterior aspect of the **intercondylar fossa** of the tibia; prevents **excessive posterior translation of the tibia on the femur**

Clinical Note: PCL injury: common mechanism of injury involves **'dashboard'** type event; an object strikes the anterior aspect of the tibia when the knee is in a flexed position, forcing the tibia to translate posteriorly relative to femur.

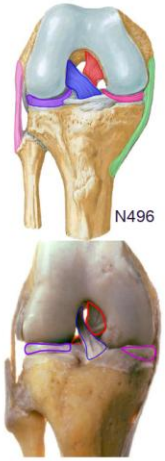
- **PCL assessment:** PCL integrity is tested via the ***posterior drawer test:*** similar to anterior drawer test, but examiner draws tibia posteriorly under the femur. Greater than normal posterior translation indicates PCL damage

Ligamentous Injuries of the Knee (ACL and PCL injuries, MCL and LCL injuries, Evaluation, treatment, and return to play decisions) ***No text notes from Dr. Briscoe**

Meniscus Anatomy and Function

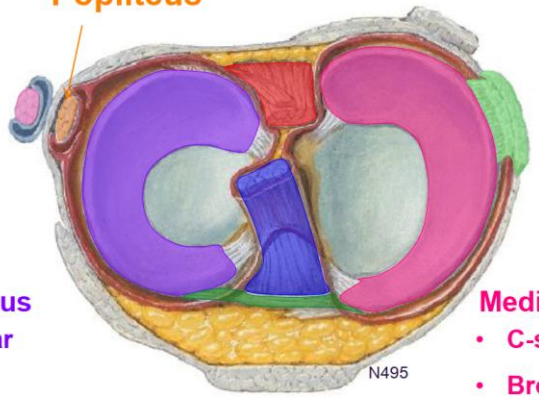
- Anatomy and function of meniscus
- Bucket-handle tears
- Locking mechanism of the meniscus

Knee Joint: Menisci



- Deepen and stabilize
- Shock Absorbers

Tendon of Popliteus



Lateral Meniscus
- Nearly circular
- Smaller
- Freely movable
- Not as Susceptible to Injury

Transverse Ligament of the Knee

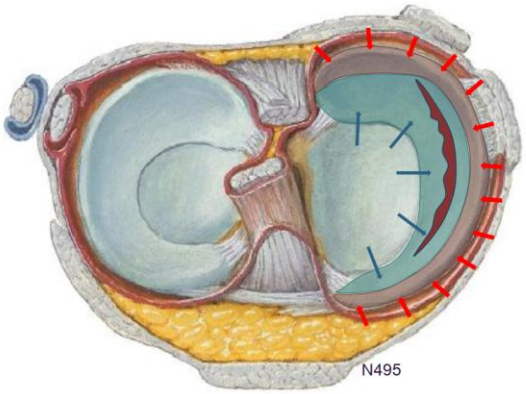
Medial Meniscus
- C-shaped
- Broader Posteriorly
- Firmly adhered to the Tibial Collateral Lig.

iii. Medial meniscus: C-shaped, broader posteriorly than anteriorly, limited mobility due to attachment to deep fibers of the TCL; increases the congruency for the articulating surface of the femoral condyles and plays a role in shock absorption

Clinical Note: the medial meniscus is the more likely to be torn/damaged than the lateral meniscus due to its **lack of mobility** and attachment to the **TCL; usually damaged** during stresses to the knee that also ***affect the TCL***, including the twisting/valgus mechanism that results in the **'Unhappy triad'** set of injuries

iv. Lateral meniscus: More **circular, smaller,** and **more mobile** of the menisci; increases the congruency for the articulating surface of the femoral condyles and plays a role in shock absorption

Medial Meniscus: Bucket-Handle Tear



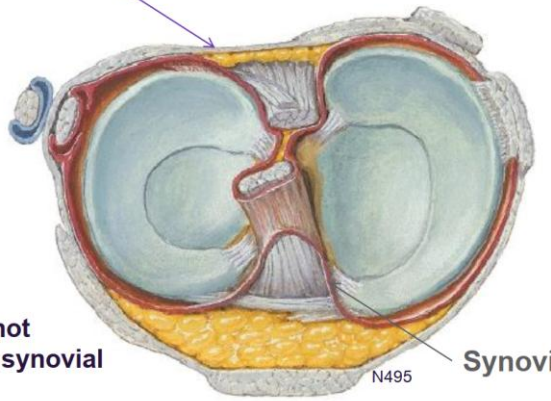
Symptoms:
- Pain
- Swelling
- Locking of the knee (due to partial detachment)

Meniscal Blood Supply:
- Peripheral meniscus richly vascularized through capsular vessels from genicular arteries
- Central meniscus is avascular; nourished by synovial fluid

Meniscal tears involving the more central portions do not heal and the fragments have to be removed (partial menisectomy)

Knee Joint: Capsule & Synovium

Fibrous Joint Capsule



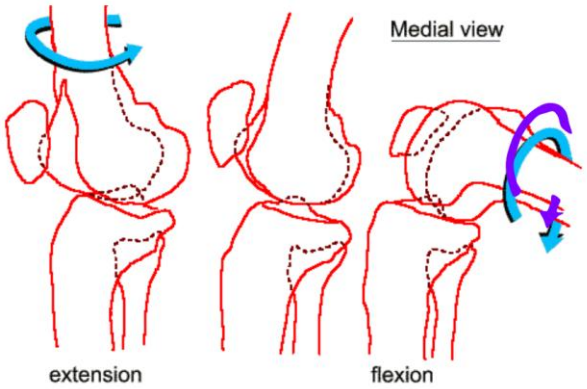
Superior View

- **ACL & PCL are not enclosed within synovial membrane**

Synovial Membrane

Knee Joint: Locking Mechanism

Femur rotates medially on tibia during the final 15 to 20° of extension



***Screw Home Mechanism**: Medial rotation maximally tightens the ligaments of the knee*



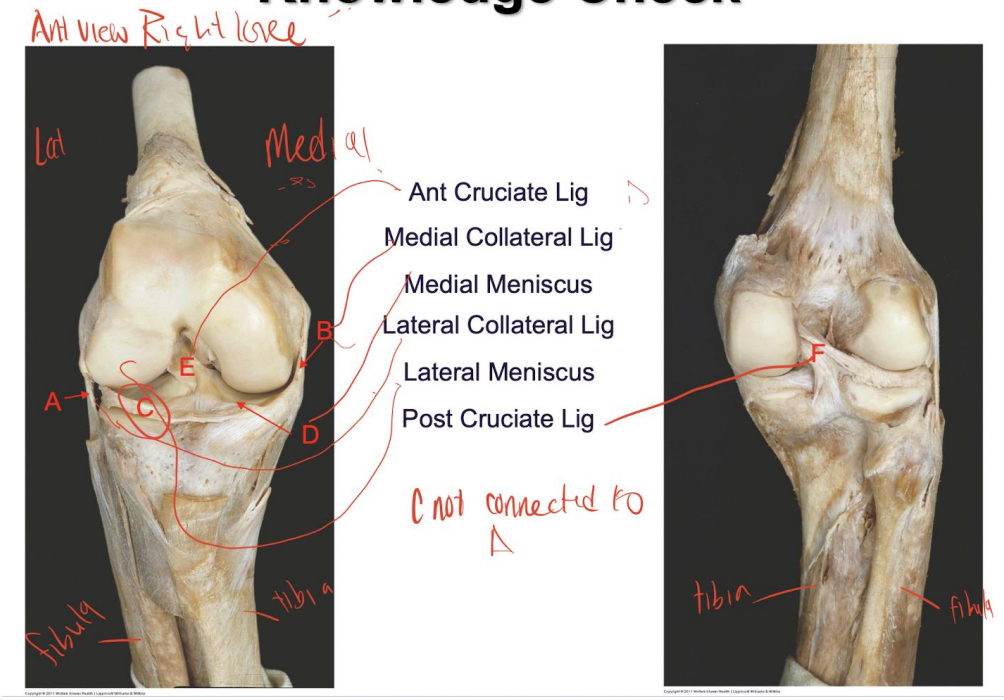
Popliteus contracts to rotate the femur 5° "unlocking the knee" for flexion to occur



Knee Extension: the shapes of condyles and menisci cause medial rotation of the femur on the tibia

Knee Flexion: Popliteus "unlocks" the knee joint by laterally rotating the femur

Knowledge Check



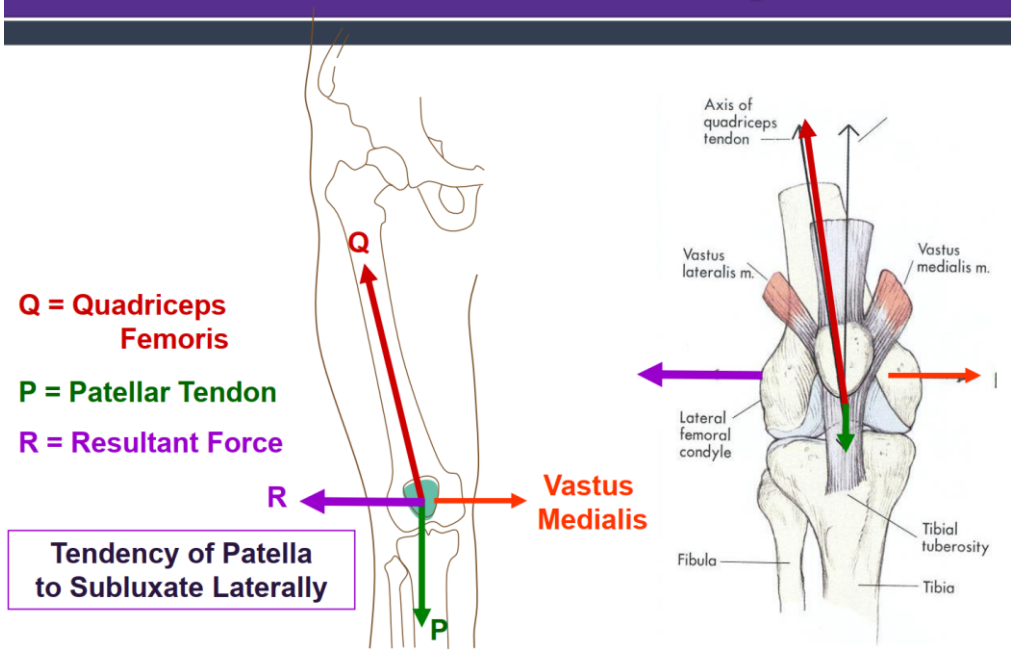
Not mentioned in text notes

Meniscus Injuries (Pathology, mechanism of injury, symptoms, Treatment of meniscal injuries, Unhappy triad) ***No text notes from Dr. Briscoe**

Anterior Knee Pain Anatomy

- Anatomy of patellofemoral joint
- Stability forces of patellofemoral joint

Knee Joint: Stability



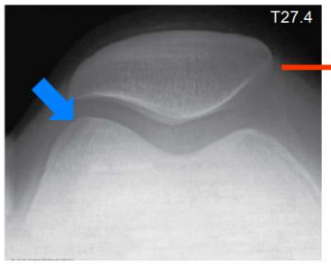
LO #2

Knee Joint: Stability

Two mechanisms counteracting the tendency for the patella to subluxate laterally during knee extension:

1. Pull by Vastus Medialis

1. Extra buttress of bone on lateral patellar surface





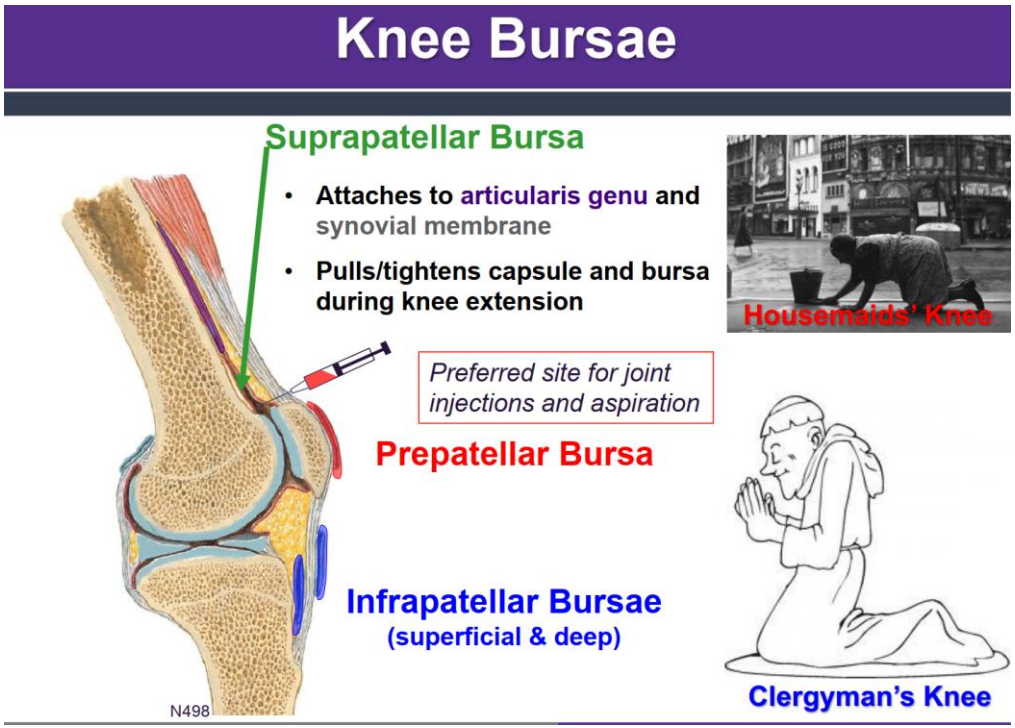
Clinical Note: **Knee Deformities:** In normal anatomical position, the femur is placed diagonally within the thigh. This creates a Q-angle (the angle between a line drawn from ASIS to patella and another line drawn through the patella and tibial tuberosity). Normal Q-angles result in normal distribution of weight between the medial and lateral femoral/tibial condyles.

- **Genu Varum (Bowlegged):** The femur is angled more vertically; thus, decreasing the Q-angle
  - Increases the weight-bearing load on the medial femoral/tibial condyles
- **Genu Valgus (Knock-kneed):** The femur is at a dramatic angle; thus, increasing the Q-angle
  - Increases the weight-bearing load on the lateral femoral/tibial condyles
- As a result of the Q-angle the quadriceps pull the patella Superolaterally, which has a tendency to subluxate the patella laterally. There are two mechanisms to resist this lateral pull on the patella: the **vastus medialis** and an extra buttress of bone on the lateral patellar surface of the femur.

Anterior Knee Pain Pathology (Patellofemoral Pain Syndrome (PFPS) - Symptoms, Presentation, Treatment) ***No text notes from Dr. Briscoe**

MISC Knee Structures

- Anatomy of Knee Bursae



Associated Structures: There are many **bursae** located around the knee that function to protectall of the tendons that are crossing the knee joint from rubbing against the bone

a. ***Prepatellar bursa:*** Located superficial to the patella allows the skin to move freely during knee movement
b. ***Infrapatellar bursa:*** Superficial and deep bursa that lie superficial and deep to the patellar ligament. Allow the skin and patellar ligament to move freely during knee movements.
c. ***Suprapatellar Bursa:*** Lies deep to the quadriceps femoris tendon and attaches to a small muscle, articularis genu, and the synovial membrane. When the quadriceps extend the knee articularis genu will pull the bursa tight to allow the quadriceps femoris tendon and patella to move freely over the femur.

Clinical Note: Irritation or swelling of a bursa due to chronic friction is known as Bursitis.

- **Housemaid's knee** is usually caused by friction between the skin and the patella, affecting the **prepatellar bursa.** Commonly seen in individuals who move around and work on their knees often.
- **Clergyman's knee** is usually caused by excessive friction between the skin and the tibial tuberosity, affecting the **superficial infrapatellar bursa.** Commonly seen in individuals who sit on their knees on hard surfaces for extended periods of time.

Innervation: **Hilton's Law:** sensory innervation to a joint is the same as the nerve supply to muscles that act upon the joint. Knee: Femoral nerve and sciatic nerve; also articular branches from obturator nerve and saphenous nerve (due to proximity)

Blood supply: ***genicular arterial anastomoses*** formed by four branches of the ***popliteal artery:***

- superior and inferior medial genicular arteries
- superior and inferior lateral genicular arteries.
  - The descending branch of the lateral circumflex femoral artery provides collateral circulation to the knee.

Note: While the ***popliteus*** does not add much posterior stability to the joint, it is important for:

a. unlocking the knee from terminal extension (initiates knee flexion); and
b. pulling the lateral meniscus posteriorly during knee flexion
   i. During natural extension the shapes of the femoral/tibial condyles and the menisci cause the femur to rotate medially on the tibia
   ii. This rotation allows the joint to be more stable, enabling 'quiet standing' (limited muscular contraction to maintain stance).
   iii. The popliteus muscle contracts to laterally rotate the femur, to 'unlock the knee', to allow flexion of the knee.

MISC Knee Pathology (IT band friction syndrome, Popliteal (Baker's) cyst) ***No text notes from Dr. Briscoe**
Knee Injection Therapies (Corticosteroids, Hyaluronic acid, Platelet Rich Plasma (PRP), Mesenchymal stem cells) ***No text notes from Dr. Briscoe**
Clinical Approach to the Knee (History, Approach to traumatic and atraumatic pain, Imaging tests) ***No text notes from Dr. Briscoe**